Quantitative Estimation of Wheat Stripe Rust Disease Index Using Unmanned Aerial Vehicle Hyperspectral Imagery and Innovative Vegetation Indices

Jie Deng, Rui Wang, Lujia Yang, Xuan Lv, Ziqian Yang, Kai Zhang, Congying Zhou, Li Pengju, Zhifang Wang, Ahsan Abdullah, and Ma Zhanhong

***Abstract*— This study aimed to identify and assess vegetation indices (VIs) and their optimal band combinations using unmanned aerial vehicle (UAV) hyperspectral imagery for the quantitative inversion of wheat stripe rust. This would offer guidance for selecting rust-resistant phenotypes and facilitate large-scale disease monitoring using aerial and spaceborne remote sensing images. The experimental design encompassed 960 wheat varieties (strains) in agricultural fields. Hyperspectral imagery was acquired at 100-m altitude during different disease stages, and disease index (DI) was investigated per plot. A custom program explored VIs with two, three, and four bands using 30 calculation methods and 3 463 790 band combinations. Regression models employed threefold cross-validation and multilayer perceptron (MLP) algorithms, with the mean R^2 value indicating optimal index and band combinations. The results revealed that the chosen VIs were effective in inverting the DI. Selected two-band VIs included MGRVI (531, 571), with $R^2 = 0.746 \pm 0.01618$; the optimal three-band VIs was ARI2 (531, 550 ± 10, 640 ± 25), with $R^2 = 0.755 \pm 0.00896$; and the best four-band VIs were DBSI (531, 551, 750, 799), with $R^2 = 0.778 \pm 0.01300$, which was comparable to full-band modeling ($R^2 = 0.775 \pm 0.01508$). The models' performance improved with an increasing number of bands in the VIs. This study demonstrated that appropriate multi-VIs modeling enhances performance compared to single-VIs modeling, e.g., six combinations of VIs achieved $R^2 = 0.790 \pm 0.01141$. These findings underscore the potential of integrating machine learning algorithms and VIs for quantifying wheat rust diseases, laying the foundation for developing airborne and spaceborne imaging sensors for large-scale wheat rust monitoring.**

***Index Terms*— Hyperspectral, machine learning, unmanned aerial vehicle (UAV), vegetation indices (VIs), wheat stripe rust.**

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I. Introduction

WHEAT stripe rust, a highly prevalent and severe disease, poses a significant threat to global wheat production safety [1]. In 2020, it was listed as a Class I Crop Pest and Disease by the Ministry of Agriculture and Rural Affairs of the People's Republic of China, endangering China's food security and causing wheat yield losses of over 40% or even total crop failure in epidemic years [2]. As a typical airborne disease, its pathogen can spread long distances through the upper atmosphere [3]. Currently, an integrated disease management strategy is primarily based on cultivating disease-resistant varieties, supplemented by chemical control and cultivation practices. Therefore, the selection of resistant cultivars and conducting short-to-medium-term monitoring and early warning are crucial for disease prevention and control [4]. Within the realm of disease-resistant breeding, advancements in high-throughput sequencing and the proliferation of breeding populations have rendered conventional manual phenotyping methods, which are labor-intensive, time-consuming, and markedly affected by investigator subjectivity, as a limiting factor for achieving efficient breeding outcomes [5]. For disease monitoring, conventional point-by-point surveys conducted by plant protection personnel struggle to achieve accurate large-scale monitoring, greatly hindering the development of disease epidemic monitoring and complicating the implementation of standardized prevention and control measures. By utilizing unmanned aerial vehicles (UAVs) equipped with spectral cameras to acquire data and achieve aerial-scale quantitative inversion of wheat stripe rust disease index (DI), it is possible not only to conduct high-throughput analysis of disease-resistant phenotypes but also to provide timely and accurate monitoring and early warning for the development of wheat stripe rust disease, which can serve as an efficient alternative to traditional, labor-intensive methods.

Wheat stripe rust, caused by the specialized form *Puccinia striiformis* f. sp. *tritici*, produces dark yellow-orange spots approximately 14 days postinfection, followed by the formation of bright yellow spore columns, arranged neatly in elongated stripes, on leaves or leaf sheaths [6]. Infected leaves gradually yellow and wither, affecting normal photosynthesis and metabolism, leading to significant changes in water

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content, pigment content, and cellular production [7], [8], [9], ultimately resulting in reduced yields or even reaped nothing. Remote sensing technology is based on the intrinsic relationship between plant characteristics and spectral reflectance features under different health conditions, enabling the inversion of physiological and biochemical information [10], [11]. Recent technological advancements have promoted the development of UAVs as platforms for crop health monitoring and phenotypic measurement [12], [13]. Currently, UAVs can carry sensors, such as RGB cameras, multispectral/hyperspectral cameras, thermal cameras, and LiDAR, which can be used to analyze and measure crop biotic/abiotic stresses, yield, and physiological/biochemical indicators [14]. UAV remote sensing, belonging to the aerial scale, allows for more extensive crop detection compared to leaf and canopy scales. In contrast to satellite remote sensing, it is less affected by cloud cover and weather and offers significantly improved resolution—hyperspectral imaging typically achieves 5-cm resolution at a 100-m flight altitude, while RGB imaging can reach the subcentimeter levels. Crucially, UAV imagery shares the same bird's-eye perspective as satellite imagery, enabling the modeling [15] and identification [16] of satellite data for more extensive and accurate inversion.

Hyperspectral cameras, with their higher spectral resolution compared to RGB and multispectral cameras, are more sensitive to minute changes in leaf pigments [17]. Their continuous spectral bands help to analyze the occurrence and development of plant disease during the entire epidemic stage [8]. However, this also considerably increases the data volume and processing complexity. Consequently, researchers typically calculate the vegetation indices (VIs) from several bands or employ radiative transfer models for inversion to simplify the derivation process and improve inversion efficiency [11]. For instance, Huang et al. [18] determined that the canopy photochemical reflectance index (PRI) was highly correlated with wheat stripe rust DI. Devadas et al. [19] employed the canopy anthocyanin reflectance index (ARI) to differentiate wheat affected by stripe rust from healthy wheat. Yuan et al. [20] utilized leaf-scale spectral characteristics to distinguish wheat powdery mildew from stripe rust. Shi et al. [21] employed leaf and canopy spectral indices to achieve the classification of winter wheat diseases, including stripe rust, aphids, and powdery mildew. Zheng et al. [22] proposed the utilization of canopy PRI (570, 525, 705) and ARI (860, 790, 750) indices for monitoring wheat stripe rust during the early-mid and mid-late stages, respectively. Shi et al. [8] introduced a shape-based wavelet-based rust spectral feature set (WRSFs) for investigating the biophysical alterations in wheat leaves following rust infection. Zheng et al. [23] developed the red edge disease stress index (REDSI) based on Sentinel-2 multispectral satellite imagery for regional-scale detection of stripe rust. Dong et al. [16] combined the plant senescence reflectance index (PSRI), red-edge vegetation stress index (RVSI), and meteorological data to identify wheat stripe rust severity levels. Numerous innovative approaches and ideas have been employed in previous studies to extract characteristic spectral indices for wheat stripe rust, yielding promising results. However, the majority of these studies were limited to leaf and canopy scales, with a scarcity of indices developed based on aerial-scale datasets. Furthermore, most studies focused on distinguishing between healthy and diseased wheat, stripe rust-infected wheat, and other diseased wheat, as well as simple severity levels, treating these as classification problems. Research employing regression algorithms for the quantitative inversion of DI remained limited. Finally, the data volume in most studies was relatively small, ranging from tens to hundreds of samples, which raises concerns about the reliability of the developed indices. Additionally, the limited range of wheat varieties investigated made it challenging to ensure the efficacy of these indices under different cultivar conditions.

Hence, it is crucial to develop more robust spectral indices based on large-scale, high-resolution hyperspectral data collected at an aerial scale for diverse wheat cultivars. This will enable quantitative inversion and facilitate high-throughput disease resistance phenotyping and precise monitoring of plant disease. The objectives of the present study were as follows.

1) To utilized 960 local wheat varieties (lines) from Henan as experimental materials and acquired high-resolution hyperspectral images at a 100-m altitude using UAVs, thereby constructing a comprehensive UAV hyperspectral dataset of wheat stripe rust.
2) To delve into a suite of methodologies that efficiently exploit hyperspectral data for the development of VIs pertinent to quantitative remote sensing. By employing machine learning regression methods, the identification of optimal VIs and band combinations will be accomplished through the exploration of over 3.46 million band combinations for each of the 30 index algorithms, using an iterative programming approach.
3) Additionally, previously published indices will be compared, providing valuable insights for crop high-throughput phenotyping research and plant disease monitoring. In conclusion, this study will construct a hyperspectral dataset at the scale of UAV aerial remote sensing to quantitatively invert the DI of wheat stripe rust, which will cover the process from initial onset to severe onset of a large number of varieties. Through the iterative exploration of VIs using traversal algorithms and machine learning methods, the study will significantly facilitate the extraction of information from hyperspectral remote sensing data. The developed methodology will hold significant value for conducting similar research and applications in various domains.

II. Materials and Methods

A. Experimental Materials

1) *Test Cultivars:* The test cultivars used in this study comprised of 960 local Henan major wheat cultivars and subsequent material lines, where each cultivar was planted in 0.75 × 3.75 m plots (two rows).
2) *Test Fungal Strains:* The fungal strains used in this study were the mixed strains of prevalent Chinese wheat stripe rust physiological races CYR32, CYR33, and CYR34. The mixed strains were provided by the Institute of Plant Protection, Gansu Academy of Agricultural Sciences,

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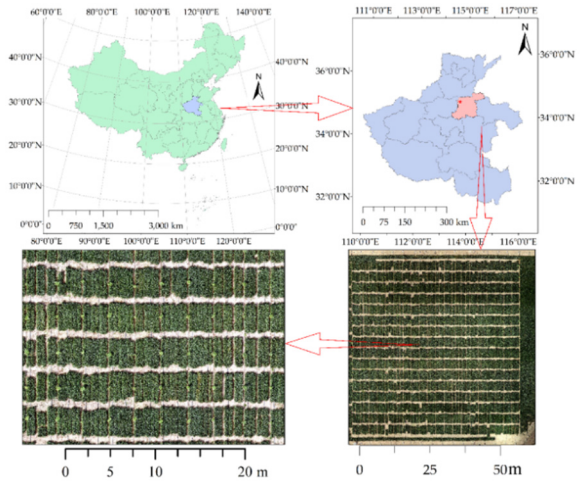


Fig. 1. Map of the study area.

Lanzhou, China and rapidly propagated by the Laboratory of Plant Disease Epidemiology, China Agricultural University, Beijing, China.

B. Study Area

This study was conducted in a field experiment at the Kaifeng Experimental Station of China Agricultural University in Xinghua Ying, Kaifeng, Henan, during 2021–2022 (Fig. 1). Seeding took place on October 15, 2021, on a total experimental area of 3600 m^2 (60 × 60 m). There were 320 plots, each measuring 3 × 3.75 m, with three cultivars planted in each plot, where each cultivar occupied two rows, totaling 960 local Henan major cultivars and their subsequent material lines. Inoculation was carried out on March 28, 2022, at 18:00 h. Before inoculation, the plots were thoroughly irrigated by sprinkler irrigation system to ensure adequate soil moisture and the presence of dense water droplets on the leaves. Wheat stripe rust spores propagated in the laboratory were removed from the refrigerator, and a 500-mg/L spore suspension was prepared using a 0.02% Tween-80 solution. The suspension was sprayed over the entire field three times using an XAG XP2020 agricultural drone (Jifei Technology Company Ltd., Guangzhou, China), flying at a height of 4.5 m, with a flow rate of 4000 mL/acre, and a speed of 5 m/s. Due to the large area, no film covering was applied after spraying; however, the soil and leaves remained moist, with many dewdrops observed on the leaves the following morning. Irrigation and inoculation were repeated on the evening of March 29, 2022, to ensure sufficient inoculation.

C. Data Collection

1) *Hyperspectral Data Collection:* Data were collected using a DJI M600 Pro drone (DJI Sciences and Technologies Company Ltd., Shenzhen, China) equipped with a Gaiasky mini2-VN (Dualix Spectral Imaging Company Ltd., Jiangsu, China) hyperspectral imaging system. The spectral range was 400–1000 nm, with a spectral resolution of 4 nm, comprising 176 spectral channels. The flight altitude was set at 100 m, resulting in a ground spatial resolution of 4–5 cm/pixel, and both the along-track and cross-track spatial overlap rates were 70%. Data were collected three times on May 3, 2022, May 13, 2022, and May 17, 2022.
2) *Disease Assessment Data Collection:* Synchronized with the collection of drone-based spectral data, surveys for DI were conducted for each cultivar in the plots, three times in total. The assessment method followed the "Technical Specification for the Monitoring of Wheat Stripe Rust" (GB/T15795—2011). The incidence and severity of diseased leaves were investigated row by row, with approximately 100 leaves per row randomly selected. Prevalence, severity, and DI calculations were conducted with the following formula:

$$F = \frac{l}{L} \times 100\% \tag{1}$$

$$D = \frac{\sum (i \times l_i)}{l} \times 100\% \tag{2}$$

$$DI = F \times D \tag{3}$$

where F represents the incidence rate, which denotes the percentage of infected leaves (l) relative to the total observed leaves (L). D represents the mean severity of the infected leaves, where i represents the severity levels, consisting of eight categories: 1%, 5%, 10%, 20%, 40%, 60%, 80%, and 100%. These values indicate the percentage of the infected area relative to the total leaf area. l_i represents the number of infected leaves corresponding to each severity level, and l represents the total number of infected leaves. DI serves as a comprehensive indicator for assessing the incidence and severity of the disease outbreak, presenting the average level of disease occurrence.

D. Data Preprocessing and Annotation

The hyperspectral images were initially processed using SpecView, lens calibration files, ground whiteboard reflectance data, and 40% reflectance gray cloth values at the same flight altitude for lens correction, reflectance correction, and atmospheric correction, yielding reflectance image data. The HiRegistrator software (Dualix Spectral Imaging Company Ltd., Jiangsu, China) was employed to perform coarse spatial correction on the hyperspectral images based on the visible light images acquired at the same time. Subsequently, Agisoft Metashape software (Agisoft LLC, St. Petersburg, Russia) was used to stitch the hyperspectral images, including aerial triangulation (Align photographs), dense point cloud generation (Build Dense Cloud), surface model creation (Build Mesh), digital elevation model generation (Build DEM), and the production of hyperspectral orthoimages (Build Orthomosaic).

Supervised learning was employed for model construction, necessitating the labeling of DI for each row of wheat in the images and the recording of unique sequence numbers (FID). Due to the large size of the image files, ArcGIS Pro 2.5 software (Esri) was used for DI annotation and the generation of annotation images of the same size. Utilizing the Groupby function in the Pandas API for Python, all point data

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were grouped by FID value, and average spectral reflectance was calculated.

E. Vegetation Index (VI) Iteration and Selection

We selected 30 common index calculation forms (Table S1) and used custom Python scripts to iteratively calculate the hyperspectral bands. The iterative wavelength range was 396–1000 nm, encompassing 176 bands, with combinations in the order of $b4 > b3 > b2 > b1$ (i.e., at 4-nm intervals) sequentially processed. For every two-band index, there were 15 400 iterations, whereas each three-band index had 893 200 iterations. Due to the large computational demands of four-band indices, iterations were performed at 8-nm intervals, resulting in 2 555 190 iterations for each four-band index. As the correlation coefficient method is not suitable for nonlinear relationships, consequently, the multilayer perceptron (MLP) regressor algorithm was utilized to model the index values and DI values directly, employing threefold cross-validation. The mean coefficient of determination (R^2) was used as a metric for evaluating and selecting the optimal index and band combinations. The detailed Python script can be obtained at the following link: https://github.com/caudjcc/Multi-threaded-Vis-traversal.git.

F. Multifeature Selection

To assess the effectiveness of multifeature modeling, feature selection was performed using the Max-Relevance and Min-Redundancy (mRMR) algorithm from the Pymrmr package [24]. Initially, we selected the top 60 band combinations with the highest R^2 values for each index type to create a feature dataset. For example, there were 12 types of two-band indices, each with 60 band combinations, resulting in a total of 720 features. Subsequently, the mRMR algorithm was applied to this dataset for further feature selection, and datasets containing 100, 24, 12, 6, and 3 features were created and modeled to observe the performance of models with varying feature counts.

To compare with the mRMR method, we also utilized the recursive feature elimination (RFE) technique from the Scikit-learn library for feature selection. In each round, the gradient boosting regressor algorithm was employed for training, and the bottom 0.05% of features, ranked by importance, were eliminated. This process continued with the updated feature sets until the target number of features was reached. Likewise, datasets containing 100, 24, 12, 6, and 3 features were created and modeled to observe the performance of models with different feature counts.

G. Model Construction and Evaluation

In this step, the feature sets selected from Section II-F were employed to construct machine-learning regression models. In total, 12 classical regression models were built: linear regression (LinearRegression), k-nearest neighbors regressor (KNNRegressor), support vector regression (SVR), ridge regression (Ridge), MLP regressor, decision tree regressor (DecisionTree), extra-trees regressor (ExtraTree), eXtreme gradient boosting regressor (XGBoost), random forest regressor (Random Forest), AdaBoost regressor (AdaBoost), gradient boosting regressor (GradientBoost), and bagging regressor (Bagging). All models were implemented using the Scikit-learn library [25] in Python. The data were standardized and used as an input for all models, which were built using a threefold cross-validation method.

Common evaluation metrics for regression models, which are used for continuous value prediction, were employed to assess the performance of these models. These metrics included the coefficient of determination (R^2), mean absolute error (MAE), mean squared error (MSE), and root MSE (RMSE)

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{4}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2} \tag{5}$$

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{6}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2}{\sum_{i=1}^{n}\left(y_i - \bar{y}\right)^2}. \tag{7}$$

Within these metrics, n denotes the number of samples, y_i represents the actual value of the ith sample, $\hat{y}_i$ signifies the predicted value of the ith sample, and $\bar{y}$ indicates the average of the actual values.

H. Comparison With Previously Published VIs Related to Wheat Stripe Rust

Huang et al. [18] discovered a high correlation between the PRI (531, 570) and the severity of wheat stripe rust. Devadas et al. [19] found that the ARI (550 and 700 nm) could distinguish between wheat affected by stripe rust and healthy wheat. Zheng et al. [22] revealed that the PRI (570, 525, 705) index had a higher classification accuracy for wheat and stripe rust during the early to mid-growth stages, while the ARI (860, 790, 750) index had the highest classification accuracy for healthy wheat and stripe rust-infected wheat during the mid- to late growth stages. In summary, this study compiled 18 indicators related to hyperspectral indices (Table S2) based on the literature and further tested their ability to estimate wheat stripe rust DI.

III. Results and Analysis

A. Full-Band Modeling Results

A total of more than 5 million spectral data were derived from three different date images, and a Python script was used to count the average reflectance of each spectral band by FID, and finally, 5760 average spectral reflectances and their corresponding condition indices were derived as the original dataset. Spearman correlation coefficients were computed between these data and the respective DI, and the data were then divided into nine categories, with each category's mean and standard deviation calculated. As illustrated

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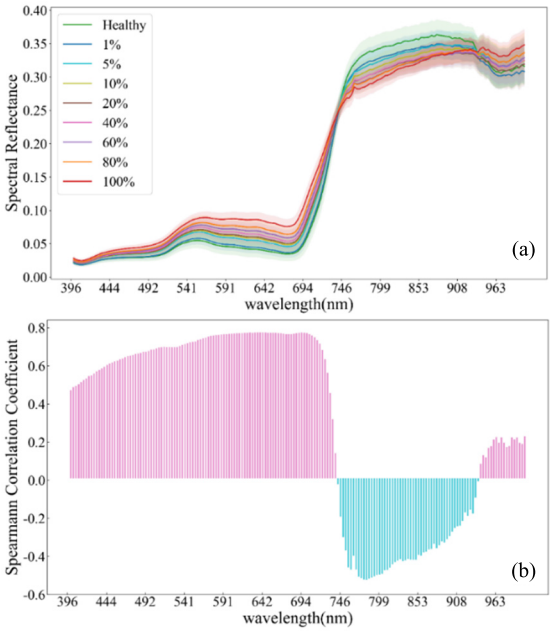


Fig. 2. (a) Spectral reflectance distribution of different DI classes and (b) Spearman's correlation coefficient between spectral reflectance and DI for each waveband.

in Fig. 2, healthy wheat exhibited lower average spectral reflectance values in the 530–690 nm range compared to diseased areas, with higher reflectance values corresponding to an increased DI. In the near-infrared band (770–870 nm), higher DI correlated with lower reflectance values. Based on this raw dataset, standardized datasets were created. Using 12 classical machine learning models, full-band modeling was conducted to estimate the DI through regression inversion, as presented in Table S3. Models, such as ridge and linear regression, performed optimally with standardized datasets, and threefold cross-validation R^2 values were 0.774 ± 0.01259 and 0.775 ± 0.01508, respectively.

B. VI Selection Results

An exhaustive evaluation of different band combinations for the VIs outlined in Section II-E was performed, selecting the band combinations with the highest mean R^2 values in threefold cross-validation. As depicted in Fig. 3 (detailed results were shown in Table S4), the inversion accuracy improved with the increase in the number of bands involved in the index: the optimal R^2 for two-band indices was 0.747 ± 0.01582, while it increased to 0.755 ± 0.00934 for three-band indices and 0.771 ± 0.01450 for four-band indices. Among the two-band indices, the best was MGRVI (531, 571), with an MLPRegressor cross-validation R^2 of 0.747 ± 0.74664. The optimal band combinations for NDVI, MSI, CIG, CSI, and IPVI were also 531 and 571 nm, with no significant differences in R^2 between them and MGRVI. This suggests that the 531- and 571-nm band combinations were critical for inverting the wheat stripe rust DI. The R^2 results for all combinations of different bands were shown in Fig. 4, indicating that for most index types, the optimal combinations concentrate in the deep blue area of band1 (520–591 nm)–band2 (541–642 nm).

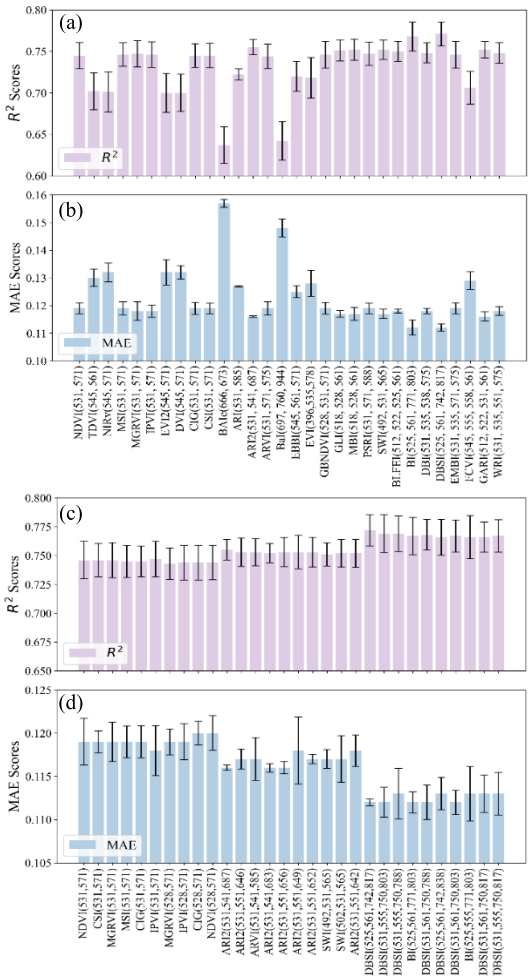


Fig. 3. Optimization of band combinations for 30 VIs through single-feature MLP regressor modeling: threefold cross-validated (a) R^2 values and (b) MAE values, as well as (c) R^2 and (d) MAE for top 10 VIs with two-, three-, and four-band combinations.

In the case of three-band indices, the best index was ARI2 (531, 541, 687), with an MLPRegressor cross-validation R^2 of 0.755 ± 0.00934. ARVI (531, 571, 575), GBNDVI (528, 531, 571), GLI (518, 528, 561), MBI (518, 528, 561), PSRI (531, 571, 588), and SWI (492, 531, 565) also demonstrated promising results. Among four-band indices, DBSI (525, 561, 742, 817) performed best, with an MLPRegressor cross-validation R^2 of 0.771 ± 0.01450. BLFEI (512, 522, 525, 561), BI (525, 561, 771, 803), DBI (531, 535, 538, 575), EMBI (531, 535, 571, 575), GARI (512, 522, 531, 561), and WRI (531, 535, 551, 575) also yielded favorable outcomes.

Figs. S1 and S2 display the scatter plots of the best band combinations for 30 index algorithms against the DI, where the relationships between various index values and the DI can be clearly observed. Notably, indices, such as MGRVI (531, 571), ARI2 (531, 541, 687), and DBSI (525, 561, 742, 817), demonstrate the most consistent trends with the DI. Data points cluster near the trend line, and the grouped error lines are relatively short, indicating a significant linear relationship. Fig. 3(c) and (d) (detailed results were shown in Table S5) presents the MLP regressor threefold cross-validation regression modeling results for the top ten indices ranked

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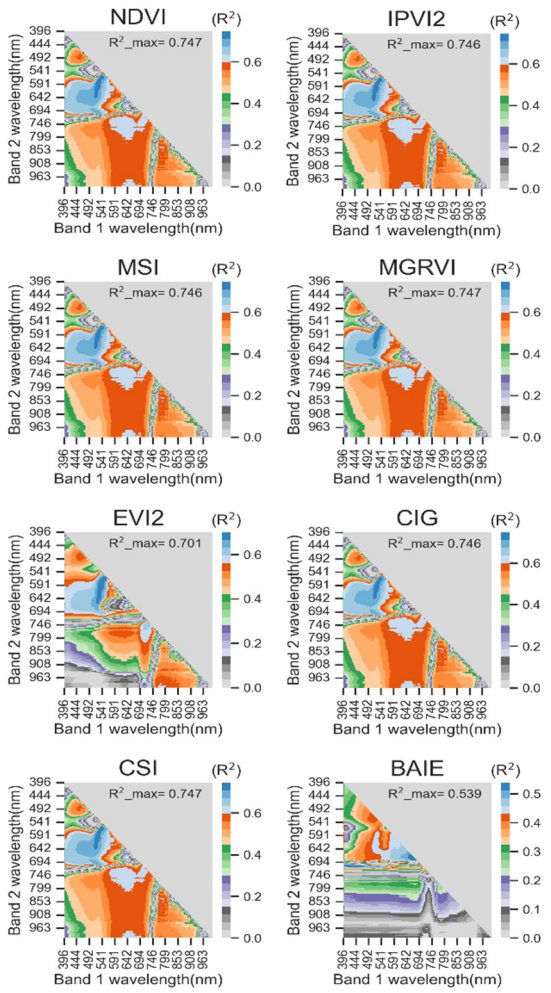


Fig. 4. Threefold cross-validation results for MLP regressor modeling of various band combinations in bi-band indices.

by R^2 scores among two-, three-, and four-band indices related to the DI. It can be observed that the best band combinations for two-band indices predominantly comprise 531 and 571 nm, with diverse index computation forms. For three-band indices, the top ten indices ranked by R^2 scores mostly employ the ARI2 computation form, whereas the top ten indices ranked by R^2 scores for four-band indices largely adopt the DBSI computation form.

C. Multifeature Index Selection Results

To determine whether utilizing multiple VIs can yield improved inversion results, the top 100 band combinations for each VI based on R^2 rankings were selected to create feature sets. These comprised 1200 features for two-band indices, 1000 features for three-band indices, and 800 features for four-band indices. The RFECV method was employed to build gradient boost models using threefold cross-validation based on different feature sets. With each iteration, the last five features were removed according to feature importance. The modeling results for different feature quantities were shown in Fig. 5. The results indicated that as the number of features increases, the R^2 value rapidly grows; however, it stabilized when the feature count reaches approximately 50.

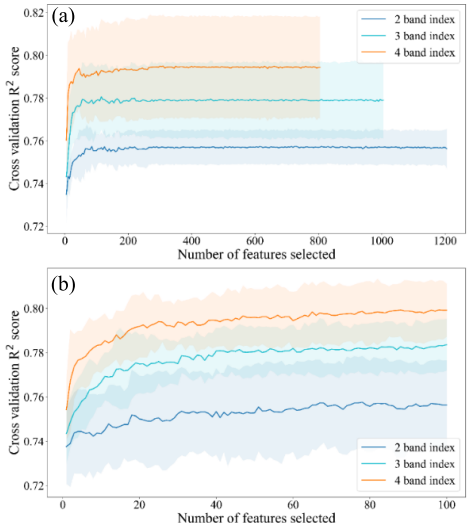


Fig. 5. Threefold cross-validation results for gradient boosting regressor modeling with varying numers of features. (a) Whole process of multifeature selection and (b) process of feature selection within 100.

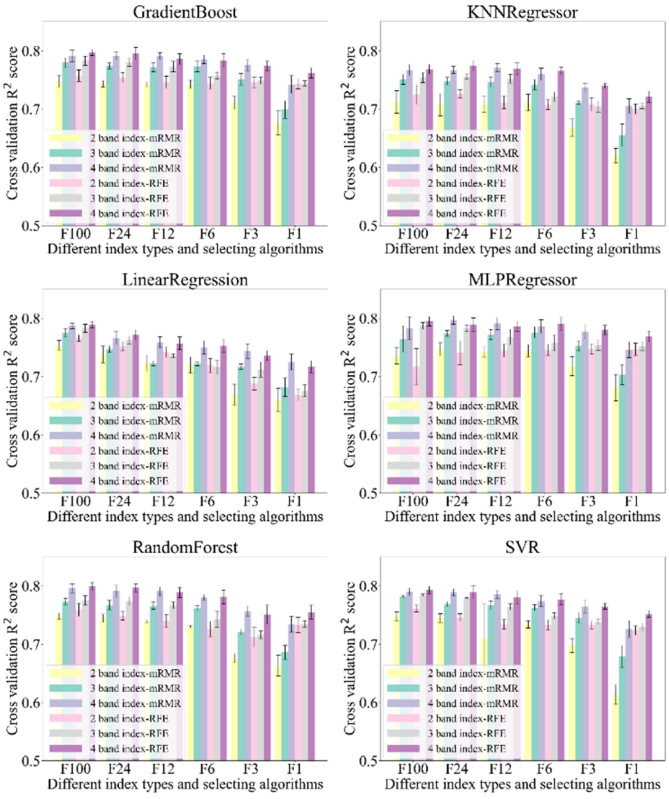


Fig. 6. R^2 results of different regression algorithms and feature sets after feature selection using mRMR and RFE algorithms (threefold cross-validation).

Utilizing the multifeature selection methods outlined in Section II-F, the different feature sets were subjected to feature selection, with the results presented in Fig. 6 (detailed results were shown in Tables S6 and S7). This study compared the modeling performance of different regression algorithms when multiple features (3, 6, 12, 24, and 100) were selected by the mRMR and RFE algorithms. The results showed that when 100 features were used, the best R^2 values for the two-, three-, and four-band feature sets selected by the mRMR

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algorithm were 0.754 ± 0.00838, 0.782 ± 0.00174, and 0.796 ± 0.00784, respectively. The RFE algorithm yields the values of 0.766 ± 0.00559, 0.787 ± 0.00556, and 0.799 ± 0.00649. When the number of features was reduced to six, the best R^2 values for the two-, three-, and four-band feature sets selected by the mRMR algorithm were 0.744 ± 0.01077, 0.776 ± 0.00940, and 0.786 ± 0.01154, respectively. The RFE algorithm yields the values of 0.746 ± 0.00979, 0.758 ± 0.01343, and 0.790 ± 0.01141.

The results indicated that the RFE algorithm performs similar to the mRMR algorithm in multifeature selection, with no significant differences within the error range. The results also revealed that modeling performance was better for four-band index feature sets than for three-band index feature sets, which in turn outperform two-band index feature sets. Furthermore, single-feature modeling results were significantly inferior to multifeature modeling results. Considering the balance between computational cost and modeling performance, selecting six or 12 features for modeling was advisable. For instance, when using the MLP regressor algorithm, the R^2 value for 100-feature modeling was 0.795 ± 0.00818, while it was 0.790 ± 0.01141 for six-feature modeling, 0.780 ± 0.00808 for three-feature modeling, and 0.769 ± 0.00927 for single-feature modeling. In comparison, the R^2 value for full-band modeling was 0.744 ± 0.02817.

TABLE I
THREEFOLD CROSS-VALIDATION RESULTS OF MLP REGRESSOR MODELING FOR RELEVANT INDICES

Index_name	Spearman_score	R^2	MAE
MSR(800,670)	-0.78	0.593±0.01779	0.160±0.00357
SIPI(800,445,680)	0.794	0.619±0.01604	0.154±0.00141
MTCI(750,710,680)	-0.779	0.578±0.01991	0.164±0.00383
PRI(531,570)	**-0.852**	**0.737±0.01565**	**0.121±0.00161**
PSRI(680,500,750)	0.802	0.640±0.01726	0.149±0.00227
TVI(750,550)	-0.705	0.477±0.00758	0.193±0.00191
MCARI(701,671,549)	0.716	0.473±0.01553	0.186±0.00072
PHRI(550,531)	0.831	0.664±0.01085	0.140±0.00084
ARI(550,700)	0.718	0.466±0.01422	0.191±0.00362
RVSI(712,752,732)	-0.67	0.430±0.00889	0.197±0.00179
YROI(611,452,550)	0.827	0.681±0.01953	0.137±0.00136
NPCI(680,430)	0.812	0.655±0.01397	0.145±0.00244
GI(554,667)	-0.75	0.574±0.01117	0.169±0.00199
TCARI(700,675,500)	0.675	0.407±0.00999	0.201±0.00136
NRI(570,670)	-0.697	0.510±0.00890	0.186±0.00288
YRI(730,419,736)	-0.009	0.009±0.00968	0.306±0.00245
PRI2(570,525,705)	**0.859**	**0.741±0.01586**	**0.120±0.00165**
ARI2(860,790,750)	-0.77	0.567±0.01781	0.166±0.00244

D. Comparison With Previously Published VIs Related to Wheat Stripe Rust

This study collated 18 indicators related to hyperspectral indices from the literature and further tested their ability to estimate the severity of wheat stripe rust. The results (Table I) indicated that the PRI2 (570, 525, 705) index had the highest correlation coefficient with the DI, and the single-band MLP regressor modeling also performed the best, followed by the PRI (531, 570) index. Fig. S3 displayed the scatter plots of the published related indices and their corresponding DI, illustrating the data distribution. The scatter plot trends of the PRI2 (570, 525, 705) and PRI (531, 570) indices were most consistent with the DI. Fig. S4 showed the data distribution of the predicted values of the validation set from the MLP regressor modeling of the published related indices and the true values of the DI. The model predictions of the PRI2 (570, 525, 705) and PRI (531, 570) indices were most consistent with the true values.

IV. DISCUSSION

In this study, we utilized high-resolution hyperspectral imagery from UAVs covering 960 wheat varieties (lines) to extract average spectral data at different disease stages. We employed custom Python scripts to exhaustively search for optimal wavelength combinations across 30 VIs, aiming to quantitatively assess wheat stripe rust DI. Moreover, we compared the results of various feature selection algorithms and machine learning regression models.

The spectral reflectance distribution for different DIs showed that the average spectral reflectance of healthy wheat in the 530–690 nm range was lower than that of the diseased areas. As the disease increased, reflectance increased because visible light solar radiation is absorbed by leaf pigments (such as chlorophyll) [26]. When wheat leaves are infected by stripe rust fungus, the spores progressively destroy the epidermal cells, leading to a reduction in chlorophyll content and water loss. Consequently, leaves gradually turn from green to yellow before wilting, lose their ability to absorb visible light, and exhibit higher reflectance than healthy plants. In the near-infrared range of 750–870 nm, the reflectance decreases as the DI increases. This is because solar radiation in the near-infrared range is reflected by leaf tissue structure and cells [27]. When plants are infected by stripe rust disease, the tissue structure and cells are damaged, resulting in a reduction in near-infrared reflectance [28], [29].

The outcomes of the two-band index selection (Fig. 3) revealed that the combination of 531- and 571-nm wavelengths was vital for the quantitative inversion of wheat stripe rust DI. This combination was optimal in various index forms, including NDVI $[(b1 - b2)/(b1 + b2)]$, MSI $(b2/b1)$, MGRVI $[(b1^2 - b2^2)/(b1^2 + b2^2)]$, IPVI $[b2/(b2 + b1)]$, CIG $[(b2/b1)-1.0]$, and CSI $(b1/b2)$. The absolute value of the Pearson correlation coefficient was 0.858, and the highest R^2 score for single-feature modeling using the MLP regressor regression algorithm with threefold cross-validation reaches 0.747 ± 0.01582. Previous research [18] has demonstrated that PRI (531, 570), i.e., NDVI (531, 570), was a robust spectral index for quantifying stripe rust DI. Our study further establishes that the combination of 531- and 571-nm wavelengths was a robust spectral index for quantifying stripe rust DI, with calculation formats that can include normalized VIs (e.g., NDVI) or ratio forms. This combination of

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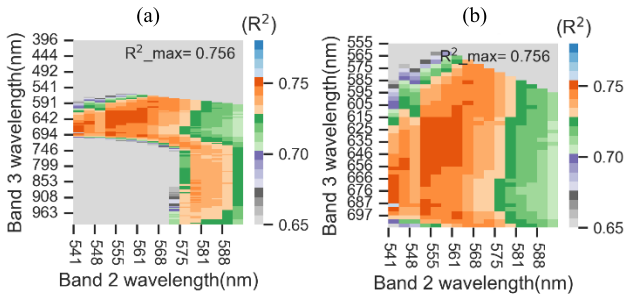


Fig. 7. MLP regressor modeling R^2 results for ARI2 index with fixed $b1$ reflectance at 531 nm and iterating through various $b2$ and $b3$ combinations. (a) Covers the spectral range from 396 nm–1000 nm of the $b3$. Label (b) Narrower segment of the $b3$ (555 nm–710 nm) to provide a better view of regions with higher scores.

wavelengths reflected the intrinsic photochemical effects in wheat leaves after infection by stripe rust fungus. Previous studies have indicated that the 531-nm reflectance index is associated with xanthophyll cycle pigments and photosynthetic radiation-use efficiency [30]. Reflectance changes at 531 nm have two sources: one near 525 nm, linked to the interconversion of xanthophyll cycle pigments, and the other near 539 nm, connected to reversible chloroplast conformational changes associated with the trans-thylakoid pH gradient [31]. Reflectance changes at 531 nm are observed in numerous diverse plant species upon increased illumination, making it a widely employed photosynthetic function index [32]. The 570-nm wavelength serves as a reference wavelength to help standardize "background" reflectance changes caused by various optical effects, such as chloroplast movement or solar angle variations [32], [33]. Shi et al. [8] demonstrated that, under stripe rust stress, anthocyanin is the first pigment to show detectable changes, with its content increase identifiable between 573 and 584 nm using wavelet features. Reflectance changes occur at wavelengths greater than 572 nm due to anthocyanin content variations, making 571 nm an ideal reference wavelength. Our study further corroborates, using aerial-scale hyperspectral data, that 571 nm is a suitable reference wavelength at the aerial scale. Combined with the 531-nm wavelength, it can robustly quantify stripe rust DI.

The results of the three-band index selection (Fig. 3) indicated that the ARI2 index form $\{b3 \times [(1/b1) - (1/b2)]\}$ offered the best inversion performance. Among the top ten three-band index R^2 score combinations, seven were of the ARI2 form, while the remainder consists of the SWI form $[b1 \times (b2 - b3)]/[(b1 + b2) \times (b2 + b3)]$ and the ARVI form $\{b3 - [b2 - 2.5 \times (b2 - b1)]\}/\{b3 + [b2 - 2.5 \times (b2 - b1)]\}$. Interestingly, all combinations within the top ten R^2 scores contained the 531-nm wavelength, further confirming that the 531-nm wavelength combination could robustly quantify stripe rust DI. Fixing $b1$ to 531 nm in the ARI2 ($b1$, $b2$, and $b3$) index form and iterating through all $b2$ and $b3$ combinations, the threefold cross-validation results were shown in Fig. 7. The optimal wavelength combinations for the ARI2 index form were concentrated at $b2 = 550 \pm 10$ nm and $b3 = 640 \pm 25$ nm, i.e., ARI2 (531, 550 ± 10, 640 ± 25). Previous research has shown that spectral absorption at 550 ± 10 nm is associated with anthocyanins [34]. When plant leaves are subjected to environmental stressors, such as strong light, UV-B radiation, low temperatures, drought, injury, and bacterial and fungal infections, they produce a large accumulation of anthocyanins to protect themselves from harmful effects. Increased anthocyanin content leads to increased spectral absorption and decreased reflectance at 550 ± 10 nm. However, between 520 and 650 nm in the green–orange range, changes in chlorophyll content also cause spectral absorption variations. With severe disease, chlorophyll content decreases dramatically, leading to increased reflectance. Consequently, spectral absorption at 550 ± 10 nm is complex, depending on both chlorophyll and anthocyanin content changes [34]. The combination of 531 and 640 ± 25 nm can effectively invert stripe rust DI.

TABLE II
MLP REGRESSOR MODELING RESULTS FOR VARIOUS DBSI INDEX WAVELENGTH COMBINATIONS USING FIVEFOLD CROSS-VALIDATION (SHOWING TOP 18 R^2 SCORES)

Index_name	R_score	R^2	MAE
DBSI(531,551,750,799)	0.871	0.778±0.01300	0.109±0.00261
DBSI(531,551,750,803)	0.868	0.774±0.01467	0.111±0.00096
DBSI(531,551,753,799)	0.869	0.773±0.01008	0.110±0.00056
DBSI(531,551,753,788)	0.870	0.773±0.01129	0.110±0.00206
DBSI(531,551,750,788)	0.869	0.773±0.01485	0.112±0.00325
DBSI(531,541,753,803)	0.863	0.771±0.01103	0.113±0.00102
DBSI(531,551,750,792)	0.865	0.771±0.01453	0.112±0.00113
DBSI(525,561,742,817)	0.868	0.771±0.01384	0.112±0.00198
DBSI(531,555,750,799)	0.867	0.772±0.01484	0.113±0.00083
DBSI(531,561,746,817)	0.867	0.771±0.01190	0.112±0.00042
DBSI(531,558,746,817)	0.865	0.771±0.01332	0.111±0.00199
DBSI(531,551,750,817)	0.866	0.771±0.01353	0.112±0.00183
DBSI(531,551,750,796)	0.865	0.771±0.01432	0.112±0.00098
DBSI(531,551,750,810)	0.866	0.770±0.01626	0.111±0.00161
DBSI(531,551,746,817)	0.864	0.771±0.01366	0.113±0.00226
DBSI(531,558,746,803)	0.863	0.770±0.01618	0.112±0.00182
DBSI(531,561,746,803)	0.865	0.770±0.01421	0.112±0.00073
DBSI(531,551,753,803)	0.866	0.771±0.01386	0.111±0.00204

Among the four-band indices, DBSI $[(b4 - b1)/(b1 + b4)] - [(b3 - b2)/(b3 + b2)]$ and BI $[(b4 + b2) - (b3 + b1)]/[(b4 + b2) + (b3 + b1)]$ exhibited the best performance, particularly the DBSI index form. The optimal wavelength combination for DBSI (525, 561, 742, 817) achieved an R^2 of 0.772 ± 0.01345. By performing a more refined iteration for DBSI (i.e., with a 4-nm wavelength interval), the top 18 MLP regressor model threefold cross-validation R^2 scores were shown in Table II. The optimal wavelength combination for the DBSI index form was DBSI (531, 551, 750, 799), with an R^2 of 0.778 ± 0.01300. In the top 18 combinations, $b1$ values were predominantly 531 nm, $b2$ values were predominantly 551 nm, $b3$ values were predominantly 750 nm, and $b4$ values were predominantly 799–811 nm. This further confirmed that 531 and 551 nm were crucial wavelengths for inverting wheat stripe rust disease, and the introduction of the 750- and 799-nm

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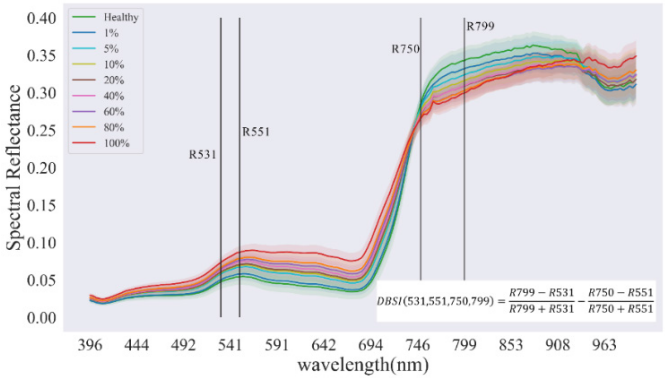


Fig. 8. Differences in average spectral reflectance among regions with varying DI values.

near-infrared bands in the DBSI four-band index form further stabilized the inversion effects of the 531- and 551-nm bands. As shown in Fig. 8, with the increased in DI, the reflectance values R531 and R551 gradually increased, with R551 exhibiting a larger increased. The reflectance values R750 and R799 gradually decreased, with R799 showing a larger decreased. Therefore, the combination of the larger changing R799 and the smaller changing R531 was used to form a normalized VI, while the combination of the smaller changing R750 and the smaller changing R551 forms another normalized VI. The final DBSI (531, 551, 750, 799) index was derived by subtracting the latter from the former.

A comparison with previously published VIs related to wheat stripe rust revealed that the inversion performance of PRI (531, 570) and PRI2 (570, 525, 705) indices was relatively satisfactory, with R^2 values of 0.737 ± 0.01565 and 0.741 ± 0.01586, respectively. These results were comparable to the two-band indices developed in this study, such as MGRVI (531, 571), and the wavelengths used were essentially consistent. However, they were inferior to the three-band index ARI2 (531, 550 ± 10, 640 ± 25) and the four-band index DBSI (531, 551, 750, 799) developed in this study. The inversion performance of the other reference indices was suboptimal, potentially because most of these indices were developed at the leaf level and canopy level, and various influencing factors at the airborne scales render them not entirely applicable.

In summary, the series of indices developed in this study, particularly DBSI (531, 551, 750, 799) with an R^2 of 0.778 ± 0.01300, achieving results consistent with full-band modeling. These indices outperformed previously published indices related to wheat stripe rust in terms of identification performance and stability. Moreover, they were developed based on airborne-scale data and show promise for application in satellite imagery to facilitate large-scale monitoring of wheat stripe rust

This study also found that the inversion performance improved as the number of bands involved in the index increased. T-SNE was employed to reduce the dimensionality of the dataset composed of the best combinations of two (531 and 571 nm), three (531, 551, and 640 nm), and four (531, 551, 750, and 799 nm) bands and visualized the results using scatter plots in 2-D (Fig. 9). The result demonstrated that the separation between the data of different DIs became more apparent with an increasing number of spectral bands.

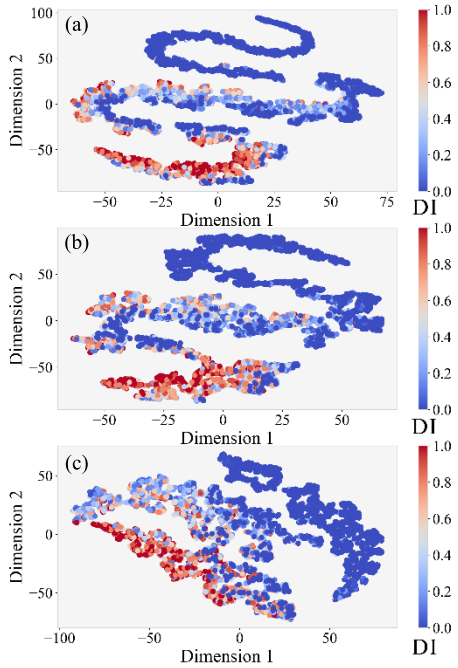


Fig. 9. T-SNE data visualization and 2-D data distribution maps obtained by applying T-SNE dimensionality reduction to spectral bands composed of (a) MGRVI (531, 571), (b) ARI2 (531, 550, 640), and (c) DBSI (531, 551, 750, 799) indices.

In addition to the development and testing of single-feature indices, this study employed mRMR and RFE algorithms to select multiple features, using a combination of indices for a more accurate inversion of wheat stripe rust DI. The results indicated that an excessive number of features did not contribute to improve the model's performance, as there was no significant difference between the outcomes of models with 24 features and those with 100 features. Utilizing multifeature modeling (e.g., three or six features) substantially enhanced the performance compared to single-feature modeling. For instance, the R^2 result of a four-band index with six features was 0.790 ± 0.01141, whereas single-feature modeling yielded an R^2 of only 0.769 ± 0.00927. Therefore, in practical applications, six- or 12-feature modeling could be employed to achieve better performance while maintaining a smaller computational load.

The feature sets selected by the mRMR and RFE algorithms were considerably inconsistent, but the final multifeature modeling results do not exhibit significant differences, which further demonstrates the more stable performance achieved by using multiple features in combination. Additionally, this study evaluated the performance of 12 classic machine learning regression algorithms, among which gradient boost and MLPRegressor exhibited consistent performance, achieving favorable results across different feature sets.

V. Conclusion

This study aimed to screen and evaluate 30 different VIs and their optimal band combinations based on airborne hyperspectral data of wheat stripe rust, with the intention to quantitatively assess wheat stripe rust disease. In this study, a series of two-band indices were identified, such as NDVI (531, 571), MGRVI (531, 571), and MSI (531, 571), which exhibited

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significant linear relationships with the DI (threefold cross-validation $R^2 = 0.746 \pm 0.01618$). The optimal three-band index was found to be ARI2 (531, 550 ± 10, 640 ± 25), with $R^2 = 0.755 \pm 0.00896$, and the optimal four-band index was DBSI (531, 551, 750, 799), with $R^2 = 0.778 \pm 0.01300$. The results demonstrated that the performance of the indices improved considerably as the number of bands used to construct the index increases, with the single-feature modeling of DBSI (531, 551, 750, 799) being comparable to the full-band modeling result ($R^2 = 0.775 \pm 0.01508$).

Furthermore, this study employed the mRMR and RFE feature selection algorithms to screen for multiple indices for combined modeling and evaluated their results. The findings indicate that multifeature modeling significantly enhances performance compared to single-feature modelings, such as the R^2 result of the four-band index with six features being 0.790 ± 0.01141, while single-feature modeling yielded an R^2 of 0.769 ± 0.00927. The results suggested that using six or 12 features in combination could yield robust modeling performance.

Overall, these findings clearly demonstrated the potential of combining machine learning algorithms and VIs for the quantitative inversion of wheat stripe rust disease. These results can serve as the foundation for the development of airborne/satellite-borne imaging sensors, enabling large-scale and efficient monitoring of wheat stripe rust.

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